

SignalForge

AI Growth Ops Engine for prop-trading firms, brokers and fintechs

Purpose, architecture, value and go-to-market | Version 1.0

Prepared for demonstration purposes

1 Executive summary

SignalForge is a ready-to-run automation product that does the daily growth and operations work of a small marketing-and-ops team for a trading brand. It is built on **n8n** (workflow automation), **Google Gemini** (AI), **Google Sheets**, **Gmail** and **Telegram**, plus a live web dashboard (“Mission Control”) backed by a managed MySQL database.

It runs four autonomous pipelines: it turns market news into ready-to-publish content, scores and answers new leads within seconds, watches a target website and its public reputation around the clock, and reports its own failures. The client configures a handful of values (brand name, site to watch, keys) and the whole system is live. The running cost of the infrastructure is close to zero because every component has a free tier.

Faster

Leads answered in seconds, content briefs ready before the team starts work.

Cheaper

Replaces repetitive manual work; software stack runs on free tiers.

Always on

Scheduled and event-driven; alerts on downtime or failure without anyone watching.

2 The problem it solves

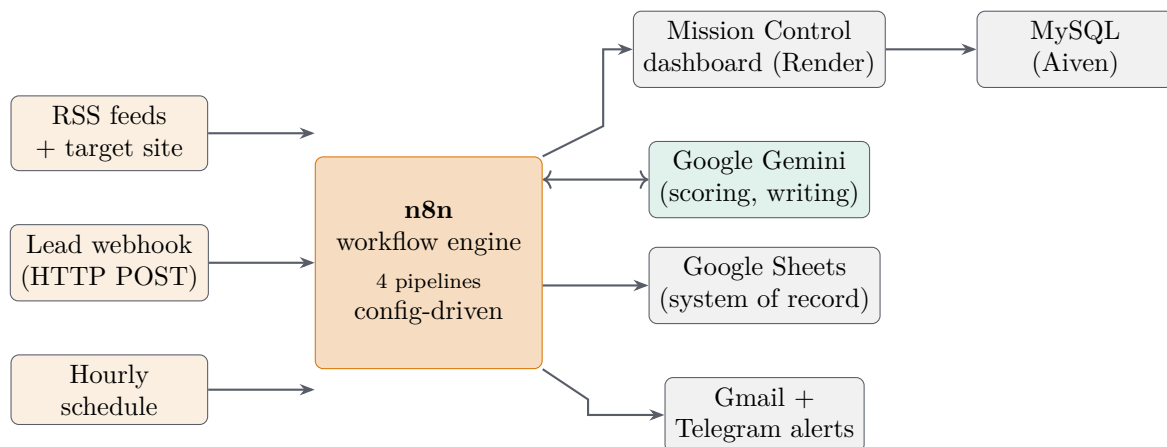
Trading brands (prop firms, forex and crypto brokers, fintech apps) compete in a fast, content-hungry, trust-sensitive market. Typical pain points:

- **Content treadmill.** Markets move daily; social posts, newsletters and blog ideas must follow, yet writing them is slow and repetitive.
- **Slow lead response.** Interested traders contact the firm at any hour. Replies that arrive hours later convert far worse than replies that arrive in minutes, and sales staff cannot tell hot leads from tyre-kickers without reading each one.
- **Blind spots.** A website outage during a promotion, or a sudden drop in public review score, is expensive and often noticed late.
- **Scattered data.** Leads, briefs and uptime checks live in inboxes and chat threads, so nobody has one view of what the automation did.

3 What SignalForge does: four pipelines

Pipeline	Trigger	What happens
A. Market intel & content	Daily 08:00, or manual button	Reads three finance RSS feeds and one target web page; Gemini writes a market brief, sentiment (risk-on / risk-off / mixed), content angles, ready-to-post X, LinkedIn and Discord copy, and a blog title with SEO keywords; a second Gemini call drafts a full blog outline. Results go to Google Sheets, e-mail, Telegram and the dashboard.
B. Lead intake & scoring	Webhook (HTTP POST)	Validates the payload, asks Gemini to score the lead 1–10, classify the trader persona, recommend a challenge size and draft a personalised reply. Hot leads (score ≥ 7) trigger an instant Telegram alert and a priority e-mail; others get a nurture e-mail. The caller receives the scoring as JSON.
C. Site & reputation monitor	Hourly	Checks that the target site is reachable, reads its public review score, logs uptime, and sends a Telegram alert and incident row if the site is down.
D. Error handler	Any failure	Catches failures in the other pipelines and sends Telegram and e-mail alerts naming the failed step and execution link.

4 Architecture



Data flow: triggers feed n8n; n8n calls Gemini for intelligence; results are written to Google Sheets (human-readable record), pushed to the dashboard database (live view) and sent to people by e-mail and Telegram.

5 Technology choices and why

Component	Role and reason for choosing it
n8n	Visual workflow engine. Each step is inspectable, retries and error routing are built in, and non-developers can adjust a flow. Hosted in n8n Cloud here; self-hostable for free for clients who need data residency.
Google Gemini API	The AI layer for scoring, summarising and writing. Chosen for its generous free tier, JSON-mode output (reliable structured results) and strong quality on short-form business writing.
Google Sheets	Zero-setup system of record that the client's team already knows how to filter, share and chart. Every action is logged as a row.
Gmail	Delivers briefs, lead replies and error reports from a normal business mailbox with OAuth sign-in (no passwords stored in the workflow).
Telegram bot	Instant push alerts to a phone at no cost; ideal for “hot lead” and “site down” moments where speed matters.
Mission Control dashboard	Hono/React web app on Render that gives a live, shareable view of briefs, leads and uptime. It is the part clients show to colleagues.
Aiven MySQL	Managed database behind the dashboard: encrypted connections, backups and a free development tier.
GitHub	Version control for the workflow export, dashboard code and documentation, so every client deployment is reproducible.

6 Benefits to the user

- **Speed to lead.** Every lead is scored, answered and routed in seconds, around the clock, so the best prospects are never left waiting.
- **Consistent content.** A daily brief and publish-ready copy arrive every morning, tied to what actually moved markets.
- **Early warning.** Downtime and reputation changes surface as phone alerts within the hour.
- **One source of truth.** Sheets for analysts, dashboard for managers, Telegram and e-mail for action.
- **Low cost and low lock-in.** Free-tier infrastructure; the workflow is a portable JSON file; services can be swapped (for example another AI model or CRM).
- **White-label by configuration.** Brand name, monitored site, review domain, recipients and keys are settings, so onboarding a new client is a configuration task, not a rebuild.
- **Transparent AI use.** Each AI output is stored beside its input, so staff can review replies before scaling automatic sending.

7 Who it is for and how to position it

7.1 Target customers

Proprietary trading (funded-trader) firms, retail forex and CFD brokers, crypto exchanges and educators, and fintech apps with a high-volume inbound lead flow and a content-driven brand.

7.2 Positioning statement

SignalForge is the always-on AI growth team for trading brands: it writes the daily market content, answers and ranks every lead in seconds, and watches your site and reputation, for the cost of a software subscription instead of a salary.

7.3 Commercial model (illustrative)

Tier	Scope	Model
Starter	Pipelines A–D, one brand, shared dashboard	Setup fee + monthly
Growth	Adds competitor watch, CRM sync, approval-before-publish	Setup + higher monthly
Scale	Multi-brand, custom pipelines, SLA and reporting	Custom retainer

Prices are deliberately left to be set per market; the margin comes from reusing one configured engine across many clients.

7.4 Demonstration flow (about 10 minutes)

1. Open the dashboard beside the n8n canvas.
2. Click *Run Demo Now*: show feeds, AI calls and the five parallel outputs lighting up.
3. Show the new dashboard entry, Google Sheet rows, e-mail and Telegram message.
4. POST a sample lead; show the score, persona and drafted reply, and the hot-lead alert.
5. Close with the configuration step: “Change three values and this runs on your brand.”

8 Roadmap

Next: competitor watch across several sites; CRM sync (HubSpot or Salesforce); human approval before auto-posting.

Then: weekly executive report; multi-brand dashboard; compliance checks for financial promotions; multi-language content.

9 Operating notes and limits

- **AI output needs review at first.** Drafted replies and posts should be sampled by a person before fully automatic sending, especially for regulated financial promotion.

- **Secrets.** API keys live in the workflow's configuration nodes for demo speed. For production, move them to n8n credentials and rotate any keys that were shared in chat.
- **Free-tier limits.** Free hosting can sleep when idle, and Gemini's free tier has daily request caps; both are fine for demos and small clients, and are upgraded as volume grows.
- **Public data only.** Scraping is limited to public pages and public review scores; always respect each site's terms.